INCH-POUND

MIL-M-28787/20A
AMEMDMENT 3
31 December 1992
SUPERSEDING
AMENDMENT 2
6 March 1986

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

INVERTER, DIGITAL, KEY CODE LDC

This amendment forms a part of Military Specification MIL-M-28787/20A(EC), dated 22 March 1979, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.2, first line: Delete "100°C" and substitute "85°C".

1.2, Output voltage, Low level: Delete "0.V" and substitute "0.4 V".

PAGE 3

3.3: Delete and substitute:

"3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.18 failures per million operating hours for class I or 0.45 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.4)."

3.4: Delete and substitute:

"3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.4)."

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PAGE 4

4.3: Delete and substitute:

"4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified."

PAGES 6, 7, AND 8

*TABLE II: Delete and substitute new table II as printed on pages 5, 6 and 7 of this amendment.

PAGE 9

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. Test conditions A, B, C, H and U are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, H, or U are part of a sequence of tests, the order of testing shall be as follows: U, A, B, C, and H where applicable, followed by the remaining tests in the sequence."

4.5.1, line five: Delete "C through N," and substitute "D through G, J through N,".

4.5.3, line three: Delete "C through R" and substitute "D through G, J through R".

4.5.3, line five: Delete "C through W" and substitute "D through G, J through T, V, and W".

6.3: Delete and substitute:

"6.3 Cross reference of classification. Military Specification MIL-M-28787/20 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing No. 10001-2833407. Part number M28787/20-1 supersedes part numbers 2833407 and M28787/20."

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Following 6.3, add new 6.4 as follows:

"6.4 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4)."

PAGE 14

FIGURE 5: Delete and substitute new figure 5 as printed on page 8 of this amendment.

PAGE 18

FIGURE 7: Delete and substitute new figure 7 as printed on page 9 of this amendment.

PAGE 21

FIGURE 8: Delete and substitute new figure 8 as printed on page 10 of this amendment.

PAGE 22

FIGURE 9: Delete and substitute new figure 9 as printed on page 11 of this amendment.

PAGE 48

*FIGURE 20, ITEM R06 box: Delete "100 Vdc ± 10%" and substitute "See note 2".

*FIGURE 20, NOTE: Delete "NOTE" and substitute "NOTES:". Number existing note as "1." and add the following:

"2. For test condition U1, use 100 Vdc ± 10%.
For test conditions U2 and U3, use 30 Vdc ± 10%."

PAGE 53

FIGURE 22: Delete and substitute new figure 22 as printed on page 12 of this amendment.

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NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Preparing activity:
Navy - NW

(Project 5963-0276)

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

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* TABLE II. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A18	VOUT (low-loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B1-B18	Input breakdown current	IV	S	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	mA
C1-C18	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D18	Output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E18	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F18	Input current, low level	VIII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G1-G18	Input current, high level	IX	X	0	30	0	40	0	40	0	40	0	40	μA
H1-H18	Output short circuit current	X	S	20	55	20	55	20	55	20	55	20	55	mA
J1	Power supply current, low level	XI	S	1	99	1	99	1	99	1	99	1	99	mA
J2	Power supply current, high level	XI	S	1	36	1	36	1	36	1	36	1	36	mA
K1-K18	Transition time (tTLH)	XII	S	4	45	4	50	4	50	4	50	4	50	ns
L1-L18	Transition time (tTHL)	XII	S	4	20	4	25	4	25	4	25	4	25	ns
M1-M18	Propagation delay time (tPHL)	XIII	S	2	19	2	22	2	22	2	22	2	22	ns
N1-N18	Propagation delay time (tPLH)	XIII	S	4	28	4	30	4	30	4	30	4	30	ns

- See footnotes at end of table.

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* TABLE II. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
P1-P6, P13-P18	Output crosstalk, low level	XIV	Q	0	970	0	970	0	970	0	970	0	970	mV
P7-P12	Output crosstalk, low level	XIV	Q	0	870	0	870	0	870	0	870	0	870	mV
P19-P36	Output crosstalk, low level	XIV	Q	0	420	0	420	0	420	0	420	0	420	mV
R1-R18	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVI	Q	0	600	0	600	0	600	0	600	0	600	mV
T	Power supply transient rise time	XVII	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
U1	Isolation													
	Initial	XVIII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
U2, U3	Isolation, unused pins													
	Initial	XVIII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	XVIII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
V1-V18	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF
W1-W36	Output impedance, power off, check	XX	Q	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	3.5	5.55	V

- See footnotes on next page.

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NOTES:

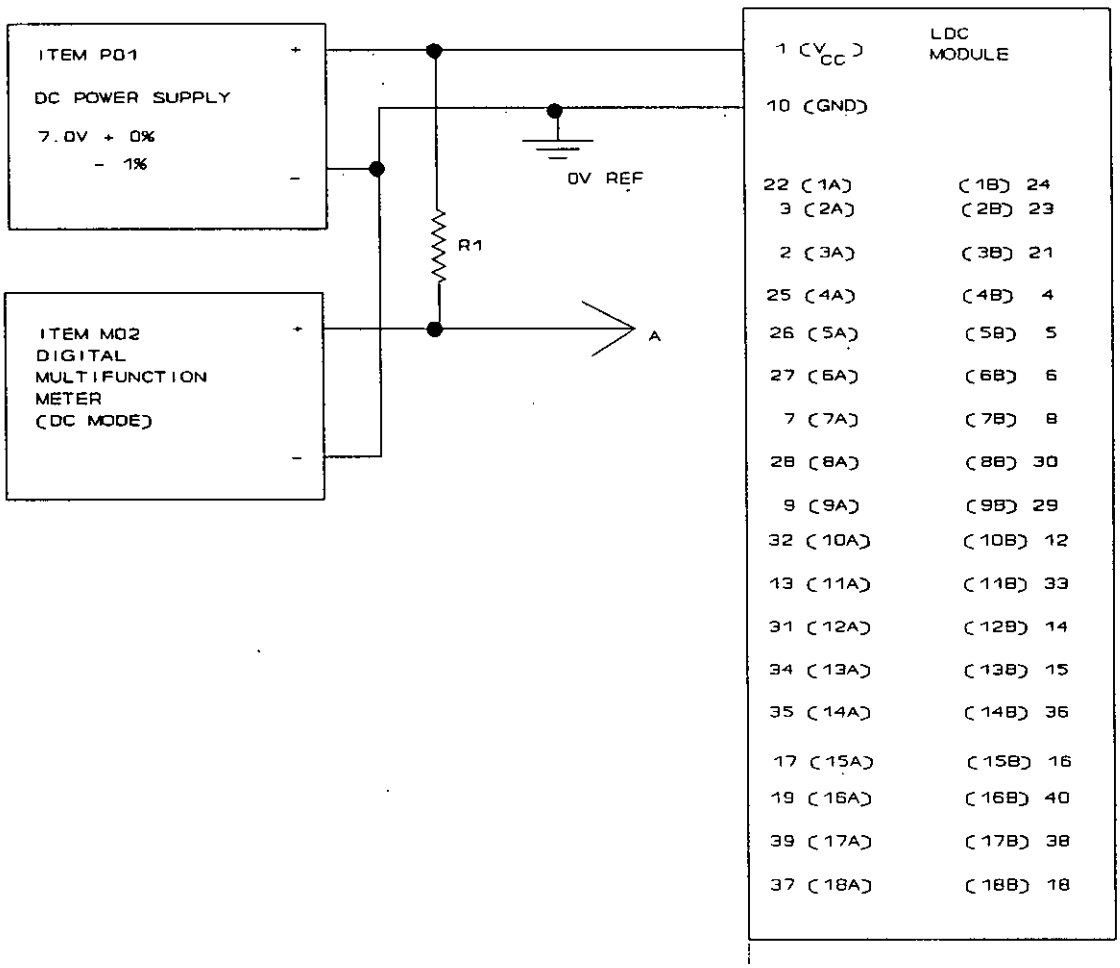
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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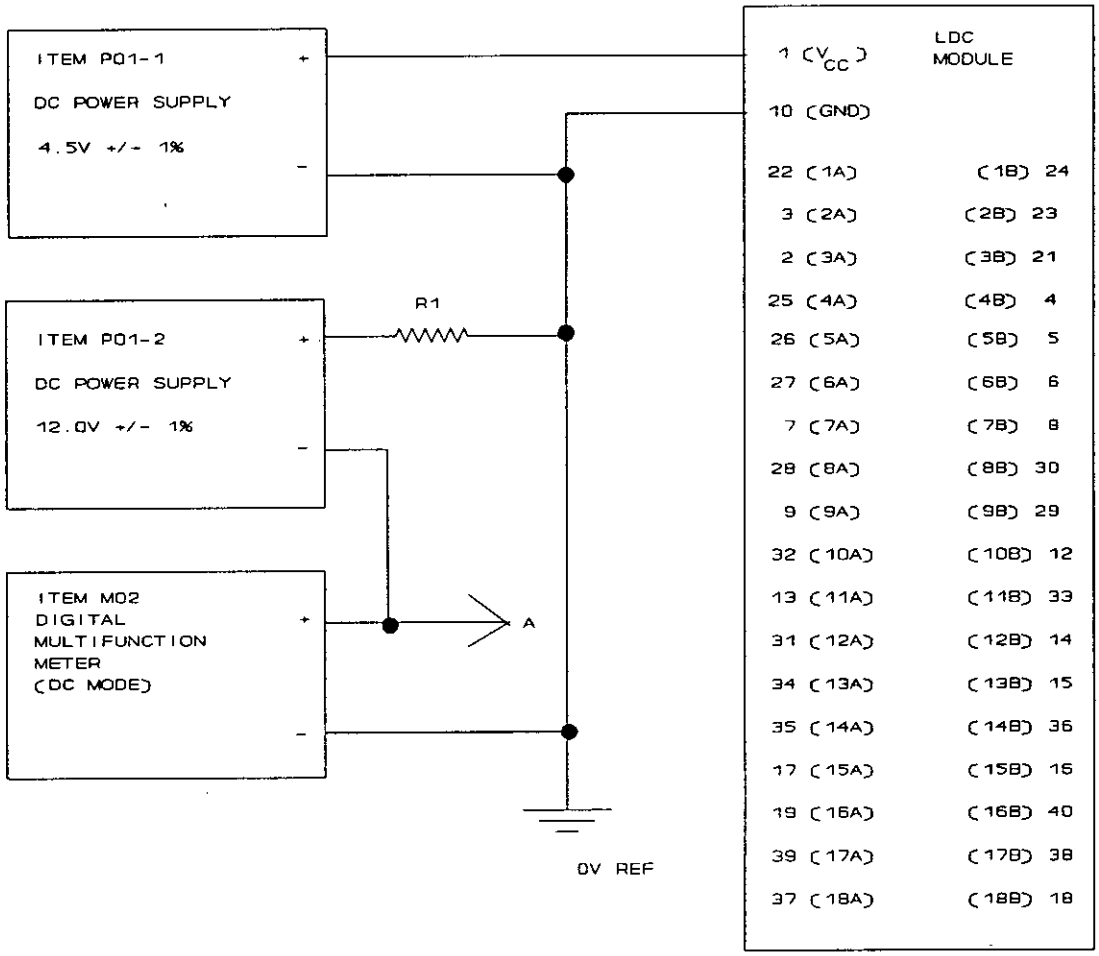
NOTES:

1. R1 = 316Ω ± 1%, 1/4 W.

2. Measurements shall be recorded after point A has been connected to the appropriate module pin for at least 5 seconds.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III.

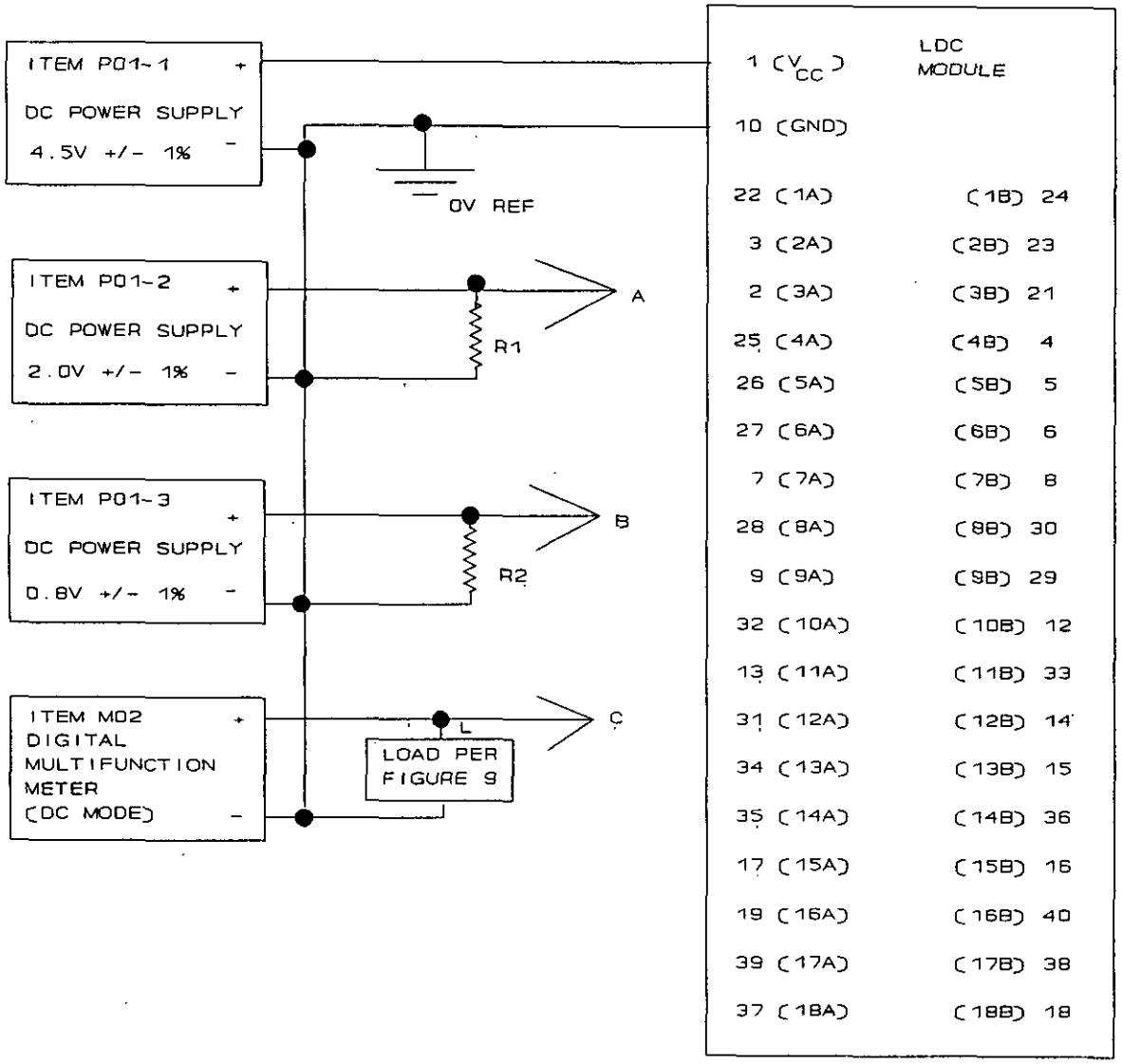
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NOTE: R1 = 877Ω ± 1%, 1/8 W.

FIGURE 7. Input clamp diode voltage test circuit for table V.

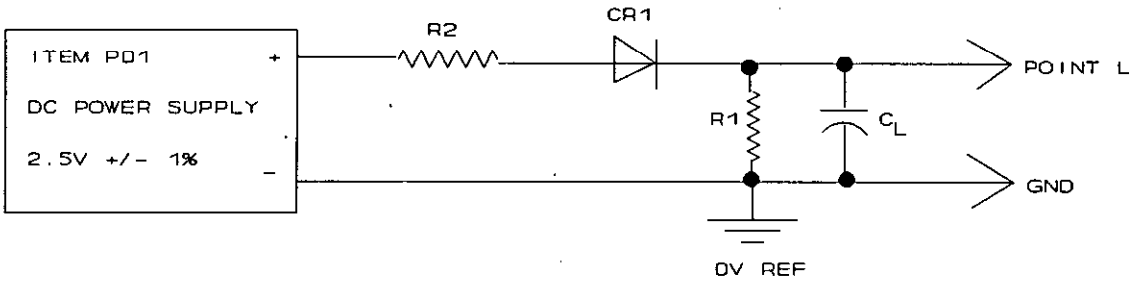
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NOTE: R1 = R2 = 51Ω ± 5%, 1 W.

FIGURE 8. Output voltage (low-level and high-level) test circuit for tables VI and VII.

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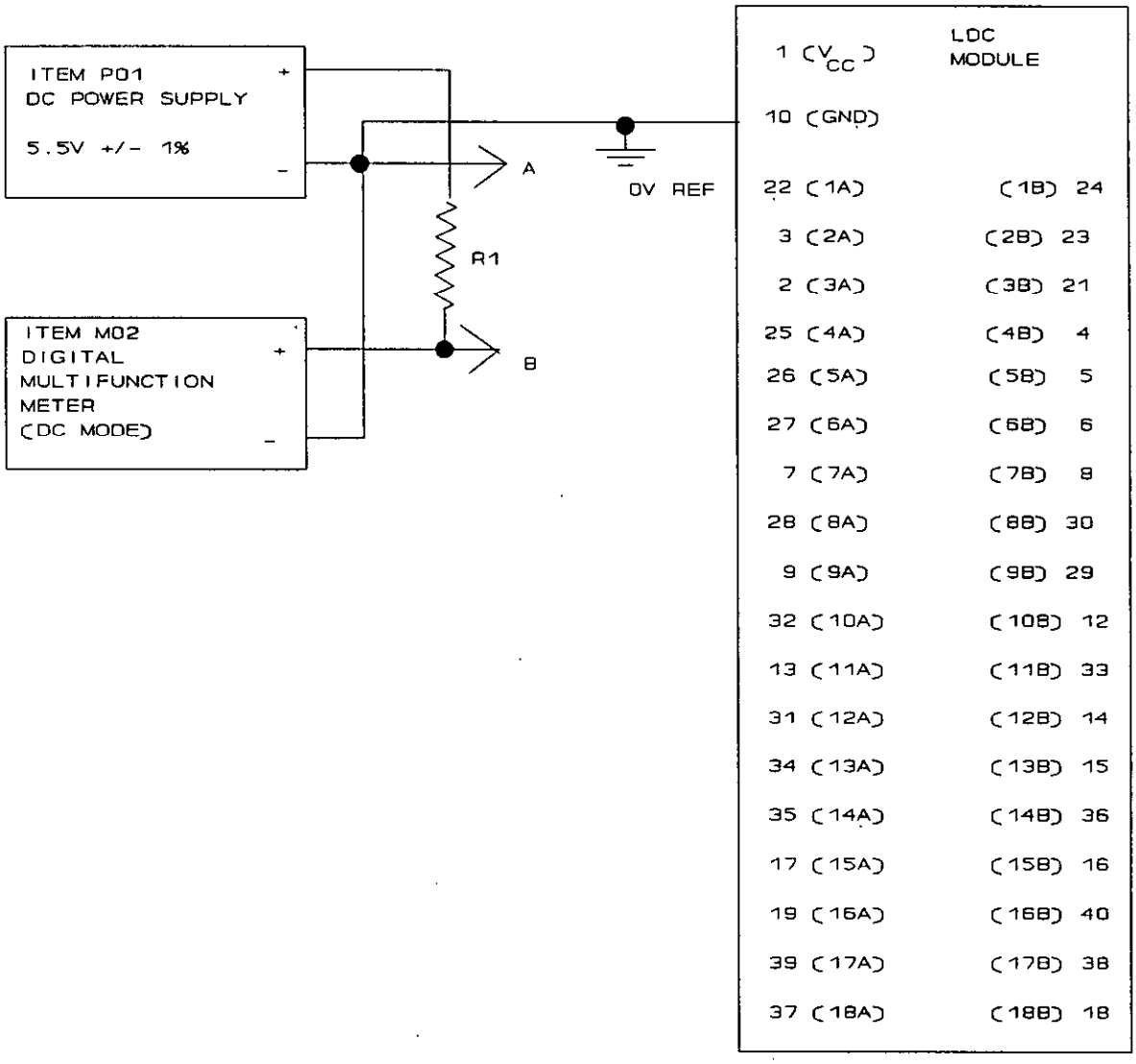


NOTES:

1. R1 = 2.43 kΩ ± 1%, 1/4 W, metal film.
2. R2 = Select value for a current flow of 16 mA ± 5% out of point L with point L held at 0.4V.
3. C_L = 100 ± 5 pF (including probe and parasitic capacitance) for test conditions K, L, M, and N.
4. C_L = 50 ± 5 pF (including probe and parasitic capacitance) for test conditions P and R.
5. CR1 = 1N4150, or equivalent.
6. C_L is not required for dc electrical tests D and E.
7. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
8. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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NOTE: R1 = 10 kΩ ± 1%, 1/8 W.

FIGURE 22. Output impedance, power off, check test circuit for table XX.

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